

Recovered Quantity of Materials from Solar PV Waste in India under Early-Loss Scenario Table 3

Waste classification	Nett PV module waste (t)	Glass (t)	Aluminium (t)	Encapsulant (t)	Silicon (t)	Precious metals (t)	Others (t)
% of waste by volume		80	10	5	4	0.5	0.5
Recovered quantity (%)		90	100	0	90	95	0
2016	2,500	1,300	250	0	90	12	0
2020P	15,000	10,300	1,500	0	540	71	0
2030P	325,000	234,000	32,500	0	11,700	1,544	0
2040P	2,300,000	1,656,000	230,000	0	82,800	10,925	0
2050P	7,500,000	5,400,000	750,000	0	270,000	35,625	0

The EU, however, has pioneered PV electronic waste (e-waste) regulations. Directive requires all producers supplying PV panels to the EU market (wherever they may be based) to finance the costs of collecting and recycling end-of-life PV panels put on the market in Europe. Lessons can be learned from the experience of the EU in creating its regulatory framework to help other countries develop locally appropriate approaches.

Action plan for future

Following actions can be taken to address the solar waste management issue:

- Policy guidelines to address the solar waste management like the Waste Electrical and Electronic Equipment (WEE) directive in European Union because the components of solar modules are not all made of biodegradable materials.
- All manufactures of solar PV modules must build in a value of recycling cost mandated by a legal framework.
- Identify causes of early loss occurring in India and measures to prevent.
- The legal framework must foster PV recycling more as a recuperate raw material for producing new modules than tagging it as a hazardous component.
- Analyse the net amount of greenhouse gases avoided in making of a new PV module against recycling waste PV modules.
- Undertake further research to make the entire recycling mechanism adequate and economically viable compared to landfilling.

• Owing to vast geographic area, portable recycling system must be developed for logistic management.

• Estimate the quantum of reproduction of PV modules from waste material.

• The disposal of recycling solar PV module must rest with the end-user as survival of module manufacturers in their lifetime may be doubtful.

• Analyse additional job opportunities generated from recycling PV modules.

Conclusions

Strong legislative laws will need to be framed by government authorities for mandatory recycling of solar energy system waste instead of landfilling, considering the massive 75,00,000 tons of waste likely to be generated until 2050. Solar module recycling presently is still not commercially viable and is a net cost for the industry. Recycling technology will have to be made economical, considering the transportation costs and minimal cost for scrap. The recovery rate should be aimed near 100% and useful to make new PV modules. The entire recycling must have zero environmental impact. Participation of producers of solar panels will be crucial to ensure the entire recycling value chain is easy. Recycling solar PV waste has potential to create enough jobs, hence must be practised. **EFY**



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